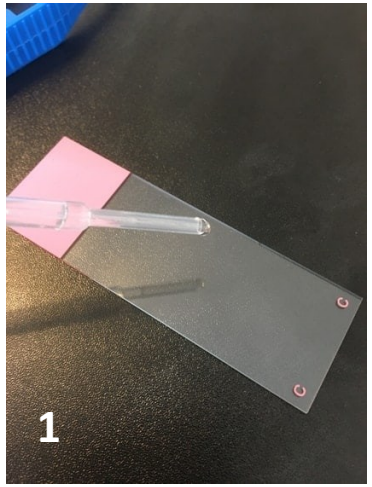
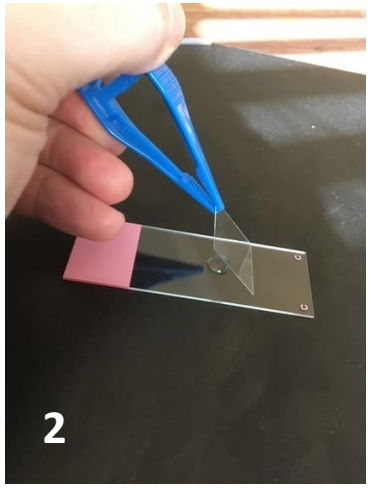
Microscopes

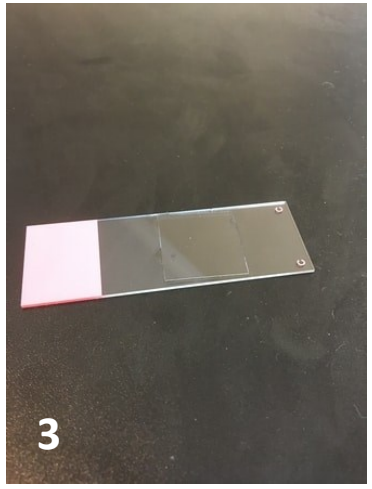
Why Use a Microscope?

- For surface water, some organisms can be indicators of the water quality
- Some organisms cannot be seen with the naked eye
- The microscope allows the scientist to correctly identify the organisms in the water

Creating a Slide







Set a slide carefully on the table.

Using a dropper, obtain a sample of water.

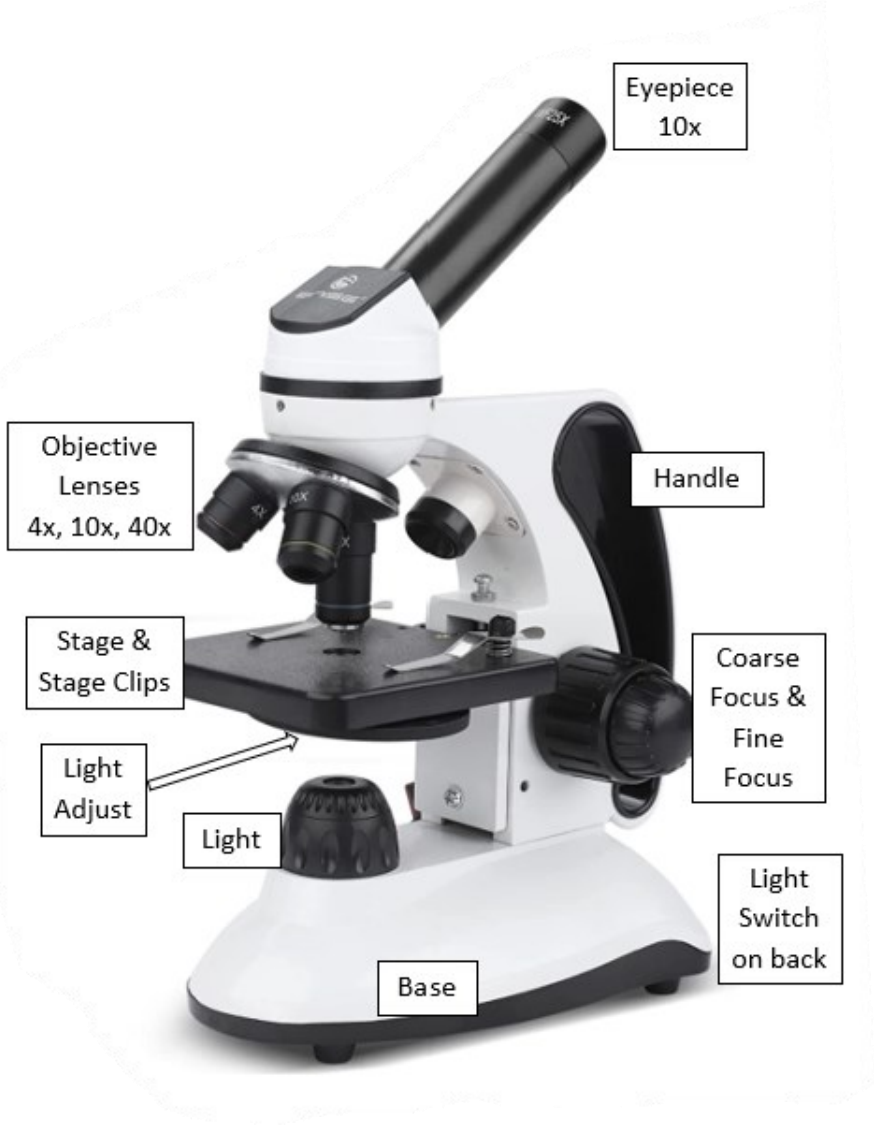
Place ONE drop in the middle of the slide.

Using tweezers, carefully place a coverslip over the drop at a 45° angle.

Gently place the coverslip down over the drop, avoiding air bubbles.

Enjoy!

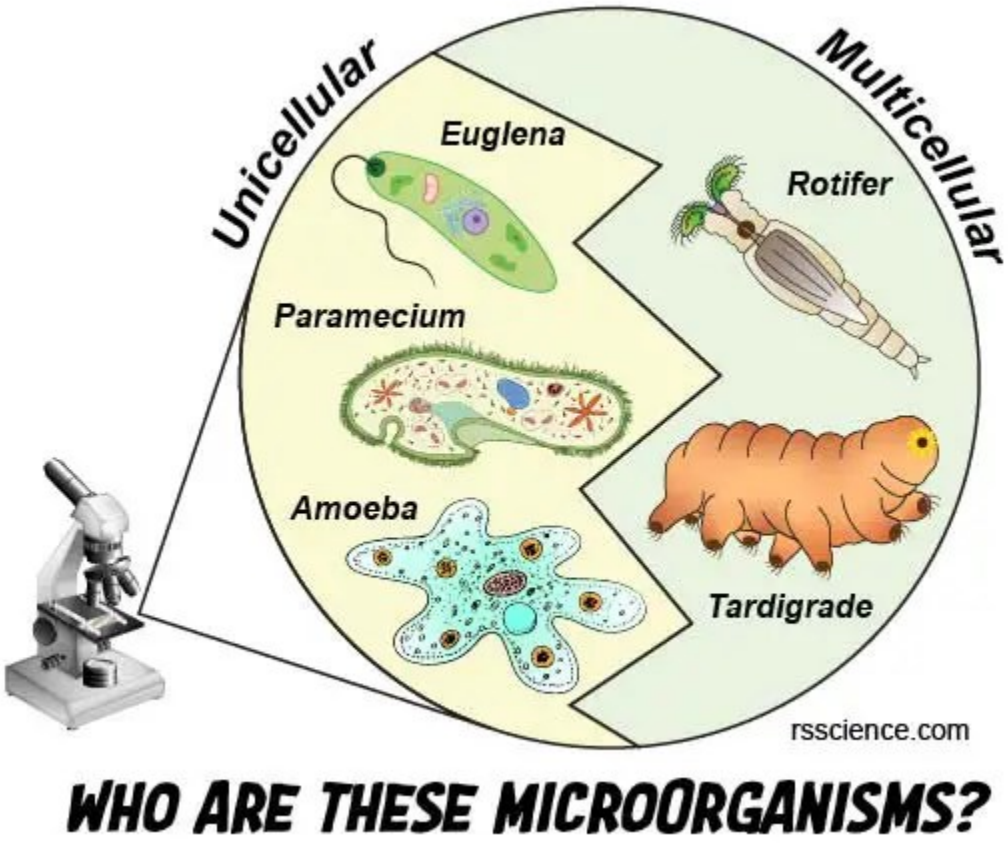
Using a Microscope

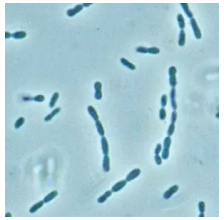
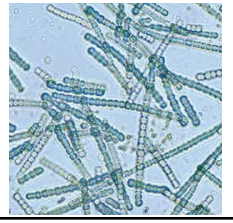
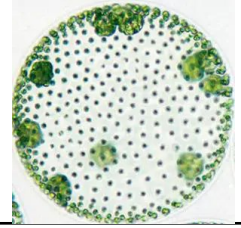
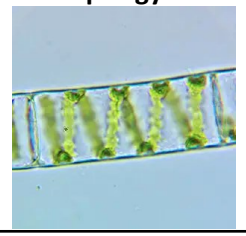
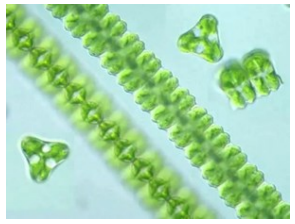
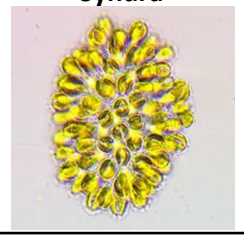
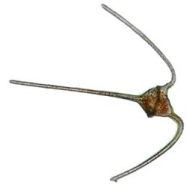
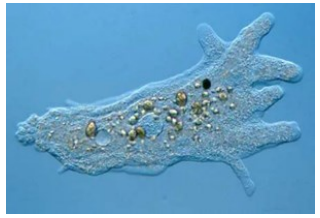
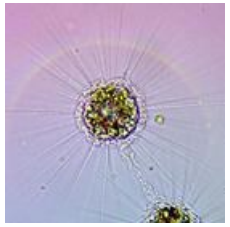
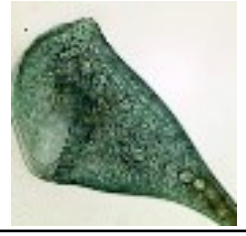
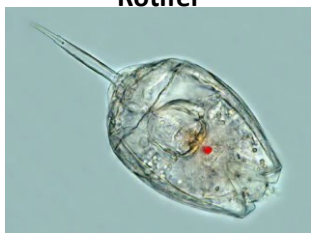
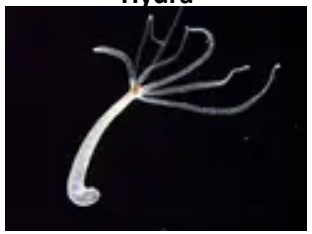
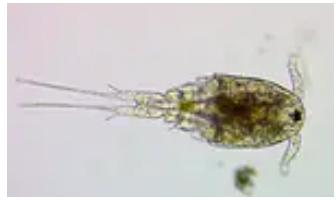


- Turn the microscope on; you will see a light
- Select the Lowest Power Objective (4x)
- Place your slide on the stage, using the clips to hold in place
- Look through the eyepiece
- Use the course focus to bring the slide into focus
- Use the fine focus knob to refine the focus
- You can increase magnification after you have the slide in focus, refocus as needed.
- You might have to move the slide around the stage to find organisms.

Oh, the Things You Will Find!

- Prokaryotic microorganisms
  - Bacteria
- Autotrophic microorganisms
  - Algae
- Protozoa
  - Amoebas
- Multicellular animals
  - Cute Tardigrades



Bacteria 	Cyanobacteria 	Chlamydomonas 	Volvox 	Pediastrum 	Spirogyra 	Closterium 
Desmidium 	Diatom 	Synura 	Euglena 	Dinoflagellate 	Amoeba 	Shelled Amoeba 
Heliozoa 	Paramecium 	Didinium 	Stentor 	Suctoria 	Vorticella 	Coleps 
Lacrymaria 	Rotifer 	Rotifer 	Hydra 	Planarian 	Nematode 	Gastrotrich 
Tardigrade 	Oligochaeta 	Daphnia 	Copepod 	Ostracod 	Amphipod 	Mosquito larva 